Logarithmic Functions

Exam code: 9MA0

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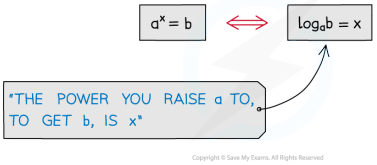
Logarithmic functions

What are logarithmic functions?



- $a^x = b$ and $\log_a b = x$ are equivalent statements
- **a > 0**
- **a** is called the base
- Every time you write a logarithm statement say to yourself what it means
  - **$\log_3 81 = 4$**
    "the power you raise 3 to, to get 81, is 4"
  - **$\log_p q = r$**
    "the power you raise **p** to, to get **q**, is **r**"

How are logarithms and powers inverses of each other?

- A logarithm is the inverse of raising to a power so we can use rules to simplify logarithmic functions

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$a^{\log_a x} = x$$

"log" AND "RAISING TO A POWER" ARE INVERSES

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How do I use the inverse property of logs and powers?

e.g. WRITE $\dfrac{2^{\log_2 81} + 3^{\log_{10} 1000}}{5^{\log_5 5}}$ IN THE FORM $\dfrac{a}{b}$

WHERE a AND b ARE INTEGERS TO BE FOUND.

"THE POWER YOU RAISE 3 TO, TO GET 81, IS 4"

$2^{\log_2 81} = 2^4 = 16$

"THE POWER YOU RAISE 10 TO, TO GET 1000, IS 3"

$3^{\log_{10} 1000} = 3^3 = 27$

$\log_5 5 = \log_5 5^1 = 1$

$5^{\log_5 5} = 5^1 = 5$

$$\therefore \frac{2^{\log_2 81} + 3^{\log_{10} 1000}}{5^{\log_5 5}} = \frac{16 + 27}{5}$$

$$= \frac{43}{5}$$

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- Recognising the rules of logarithms allows expressions to be simplified

e.g. WITHOUT USING A CALCULATOR,
EVALUATE $\log_2 64\sqrt{2}$

RECOGNISE 64 AND $\sqrt{2}$ ARE POWERS OF 2

"THE POWER YOU RAISE 2 TO, TO GET $64\sqrt{2}$, IS x"

LET $x = \log_2 64\sqrt{2}$

REWRITE THE STATEMENT

$2^x = 64\sqrt{2}$

$2^x = 2^6 \times 2^{\frac{1}{2}}$

USING LAWS OF INDICES

$2^x = 2^{6+\frac{1}{2}}$

$2^x = 2^{\frac{13}{2}}$

$x = \frac{13}{2}$

$\therefore \log_2 64\sqrt{2} = \frac{13}{2}$

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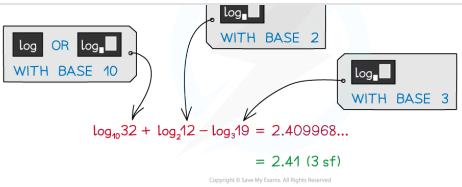
- Recognition of common powers helps in simple cases
  - Powers of 2: $2^0 = 1, 2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, \ldots$
  - Powers of 3: $3^0 = 1, 3^1 = 3, 3^2 = 9, 3^3 = 27, 3^4 = 81, \ldots$
  - The first few powers of 4, 5 and 10 should also be familiar

  For more awkward cases a calculator is needed



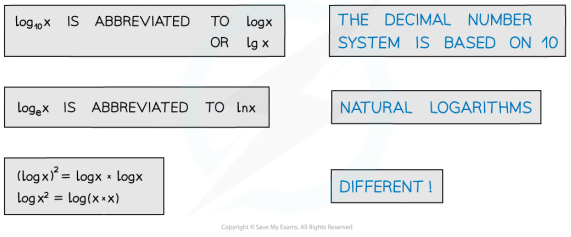
- Calculators can have, possibly, three different logarithm buttons

- This button allows you to type in any number for the base

- Natural logarithms (see "e")

- Shortcut for base 10 although SHIFT button needed
- Before calculators, logarithmic values had to be looked up in printed tables

What notation is used with logarithms?



- 10 is a common base
  - **$\log_{10} x$** is abbreviated to **log x** or **lg x**
- The value **e** is another common base
  - **$\log_e x$** is abbreviated to **ln x**

Worked Example

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